

ROSEBANK INTERNATIONAL

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Evaluating and justifying choice of dataset

Introduction

The selected dataset is the GoodBooks dataset, containing information on 10,000 books and 23 variables. The dataset includes information relating to book characteristics, authors, publication years, languages, reader ratings and reader engagement. Variables include books_count, authors, original_publication_year, language_code, average_rating, ratings_count, work_ratings_count and work_text_reviews_count.

The purpose of this project is to develop a Logistic Regression classification algorithm from scratch that predicts whether a book will be a bestseller. Since the dataset does not contain an explicit variable identifying whether a book was commercially classified as a bestseller, a proxy classification must be created. For this analysis, books in the top 20% of the dataset according to ratings_count will be classified as "bestsellers", while the remaining 80% will be classified as "non-bestsellers".

This is a reasonable proxy because a very high number of Goodreads ratings represents substantial reader exposure and engagement. However, ratings_count should not be interpreted as actual book sales. The resulting model should therefore be described as predicting high-reader-engagement/bestseller-proxy status, rather than predicting actual commercial sales.

The dataset is suitable because it is sufficiently large for classification, containing 10,000 observations, and includes several potentially useful predictors of book popularity. However, the absence of a genuine sales variable is an important limitation. Goodreads ratings measure reader engagement rather than copies sold, meaning that the classification target is an approximation rather than a direct measure of commercial success. This distinction is important in practice: a publisher could use the model as an exploratory indicator of likely high reader engagement, but it should not use the predictions as definitive evidence that a book will achieve high sales.

Planning the Analysis

Stage	Spark use	Purpose
EDA	PySpark DataFrame + Spark SQL	Understand and clean the data
Target creation	Spark	Create bestseller/non-bestseller classes
Feature selection	Spark	Identify useful predictors
Feature preparation	Spark	Transform and prepare predictors
Model training	NumPy	Implement Logistic Regression from scratch
Model prediction	NumPy	Generate probabilities/classes
Model evaluation	Spark	Analyse predictions and classification errors
Retraining	NumPy	Improve the algorithm
Final evaluation	Spark	Compare models
Report		Explain findings and communicate results

The analysis is deliberately structured so that the model is not trained before the quality and suitability of the data have been established. This is important because a high-performing model built from poorly prepared or inappropriate variables could produce misleading conclusions. The plan also recognises that accuracy alone is insufficient because the bestseller class represents only 20% of the dataset. Precision, recall and F1-score will therefore be used to determine whether the model can actually identify high-engagement books rather than simply predicting the majority class.

Conducting Analysis

Exploratory data analysis (EDA)

This section will detail the Exploratory Data Analysis (EDA) conducted on the books dataset to uncover structural characteristics, data distributions, and initial patterns. It will include descriptive statistics and visualizations to examine key metrics such as average ratings, ratings counts, and publication years while identifying any missing values or anomalies. Additionally, this analysis will guide the target variable engineering for best-seller classification and verify that the data meets the necessary prerequisites for training logistic regression model from scratch.

```
Number of records: 10000
Number of columns: 23
```

The dataset contains 10,000 observations and 23 columns, which is adequate for developing and evaluating a binary classification model. The relatively large number of observations provides enough examples of both high- and lower-engagement books for the algorithm to learn patterns.

Spark schema was displayed, the schema allows me to identify which variables Spark has interpreted as numerical and which are categorical. This is important because Logistic Regression requires numerical predictors. The schema therefore informs later feature-selection and transformation decisions.

Descriptive statistics:

summary	average_rating	ratings_count	work_ratings_count	work_text_reviews_count
count	9998	9999	9999	9999
mean	4.002162432486497 5	54003.90110111011	59692.25862586259	2922.014901490149
stddev	0.2544444289254295	157377.7001999715	167811.44396015457	6127.088904232268
min	2.47	4.11	5510.0	3.0
max	4.82	4780653.0	4942365.0	155254.0

The descriptive statistics show that reader engagement varies substantially between books. In particular, the number of ratings has a much wider range than the average rating. This indicates that Goodbooks users differentiate much more strongly in terms of which books they engage with than in the average score they give those books.

Missing values

```
+------------------------+--------------+
|Column                  |Missing_Values|
+------------------------+--------------+
|language_code           |1084          |
|isbn                    |700           |
|isbn13                  |585           |
|original_title          |585           |
|original_publication_year|21           |
|average_rating          |2             |
|work_ratings_count      |1             |
|ratings_count           |1             |
|work_text_reviews_count |1             |
|authors                 |0             |
|title                   |0             |
|best_book_id            |0             |
|ratings_5               |0             |
|work_id                 |0             |
|ratings_3               |0             |
|books_count             |0             |
|image_url               |0             |
|ratings_1               |0             |
|small_image_url         |0             |
|id                      |0             |
+------------------------+--------------+
only showing top 20 rows
```

I removed "isbn", "isbn13", "original_title", and "language_code" as they are not material or necessary for the purpose of project.

Feature Selection

The final model used books_count, original_publication_year, average_rating and author_book_count. The selection was guided by analytical relevance, avoidance of target leakage and the correlation analysis conducted using Spark. Variables directly responsible for defining the target were excluded. This was an important methodological decision because including ratings_count would result in data leakage and would make the model appear artificially accurate without genuinely learning useful predictive relationships.

The relatively modest relationships between the selected features and the target also provide an important explanation for the model's limitations. The model is attempting to predict a complex form of reader engagement using only a small number of book characteristics. Bestseller-like performance is likely influenced by many factors that are not represented in the selected variables, meaning that even a correctly implemented Logistic Regression model may have limited predictive power.

Selecting the final features

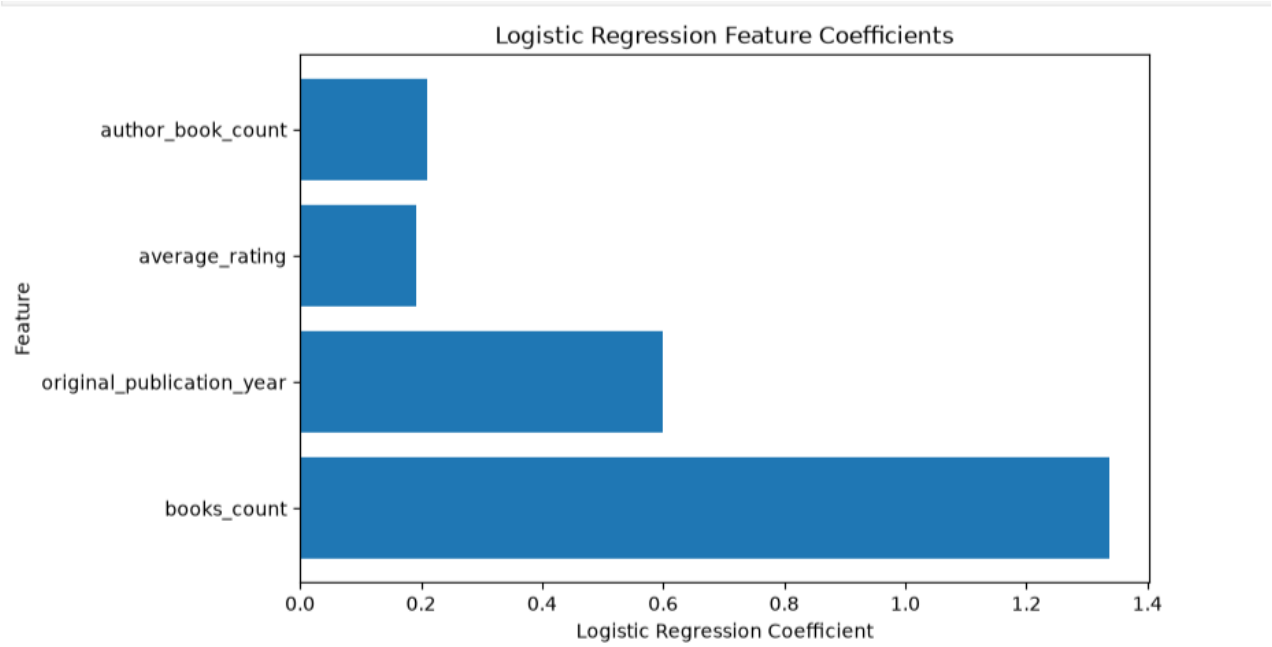
```
selected_columns = [
    "books_count",
    "original_publication_year",
    "average_rating",
    "author_book_count",
    "bestseller"
]
```

Training Model

The Logistic Regression model was trained using a manually implemented gradient-descent algorithm. The model estimated the probability that each book belonged to the positive bestseller-proxy class. A probability threshold of 0.50 was initially used to convert probabilities into binary predictions. The training process demonstrates an understanding of the mathematical mechanism underlying Logistic Regression, including the sigmoid function, gradient calculation and iterative weight optimisation.

The initial results demonstrated that the algorithm was capable of learning a meaningful classification boundary, but the low recall indicated that the default training configuration did not adequately identify the minority positive class. This finding justified the subsequent retraining stage rather than accepting the initial model as the final solution.

```
def sigmoid(z):
    z = np.clip(z, -500, 500)
    return 1 / (1 + np.exp(-z))
```



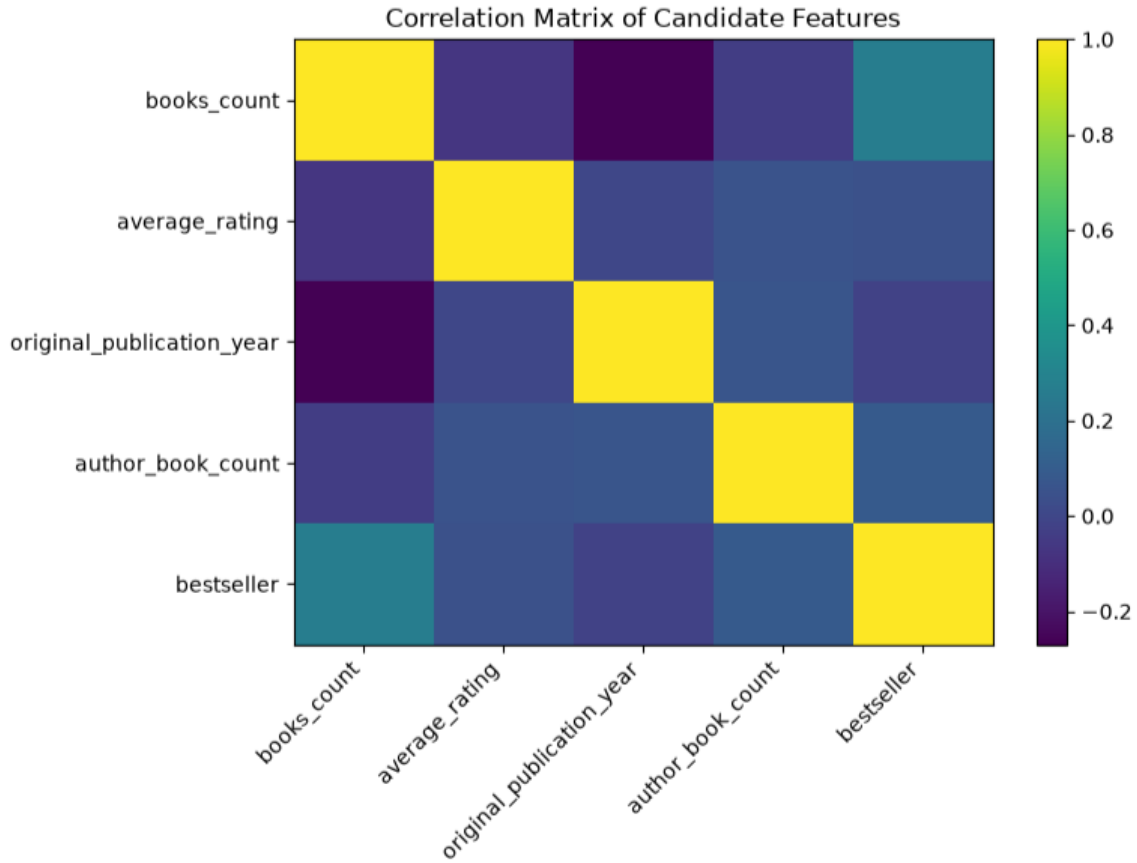
Evaluating Model

Interpreting and Evaluating the Model

The original model achieved 80.4% accuracy, which initially appears strong. However, the much lower recall of 9.8% indicates that the model identified only a small proportion of the actual bestseller-proxy books. This demonstrates why accuracy should not be interpreted in isolation. With an imbalanced target, a model can achieve relatively high accuracy by favouring the majority class. From a practical perspective, such a model would not be sufficiently useful if the objective is to identify potential high-engagement books.

The original model's precision of 61.5% indicates that predictions of the positive class were relatively selective. However, this selectivity came at the cost of missing many positive cases. The F1-score of 16.9% confirms that there was a poor balance between precision and recall. Consequently, the original model's apparent strength in accuracy masks an important weakness in its ability to identify the minority class.

```
Accuracy : 0.8041082164328658
Precision: 0.6153846153846154
Recall   : 0.09852216748768473
F1 Score : 0.16985138004246284
```



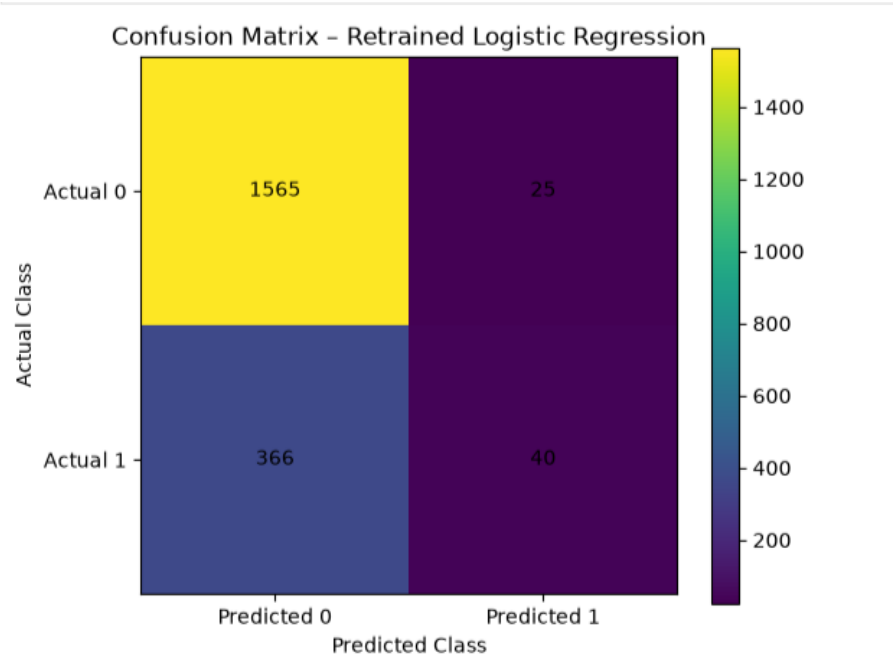
Retraining model with different Parameters

The model was retrained using class weighting to address the imbalance between the positive and negative classes. A positive-class weight of 4 was introduced so that errors involving bestseller-proxy books had greater influence during gradient descent. This modification was designed specifically to address the major weakness identified in the original model, its very low recall.

The retrained model produced an accuracy of 75.3%, which represents a reduction from the original 81.2%. However, this reduction should not automatically be interpreted as deterioration. Precision decreased from 61.5% to 42.2%, indicating that the model generated more false-positive predictions. At the same time, recall increased dramatically from 9.8% to 58.3%, while F1 increased from 16.9% to 49%. The improvement in recall means that the retrained model identified a substantially larger proportion of the books belonging to the positive class.

The comparison demonstrates an important machine-learning principle, the best model is not necessarily the model with the highest accuracy. In this case, the retrained model sacrifices some accuracy and precision in exchange for a substantial improvement in recall and F1-score. Since identifying the positive class is central to the purpose of the analysis, the retrained model provides a more appropriate balance of performance measures.

```
Retrained Accuracy : 0.753006012024048
Retrained Precision: 0.42245989304812837
Retrained Recall   : 0.583743842364532
Retrained F1 Score : 0.49017580144777667
```



Conclusion

This analysis successfully implemented a Logistic Regression algorithm from scratch using a Goodbooks book dataset, while Apache Spark was used throughout the data preparation, exploratory analysis, feature investigation and model evaluation stages. The analysis demonstrated the importance of data cleaning, appropriate feature selection, standardisation, gradient-descent optimisation and evaluation using multiple performance metrics.

The original model achieved an accuracy of 80.4%, but its recall of only 9.8% demonstrated that it was ineffective at identifying a large proportion of bestseller-proxy books. Retraining the model with a positive-class weight improved recall substantially to 58.3% and increased the F1-score from 16.9% to 49%. Although accuracy declined to 75.3% and precision declined to 42.2%, the retrained model provides a more appropriate balance for the objective of identifying potential high-engagement books.

The findings demonstrate that model performance should not be judged by accuracy alone, particularly when the target classes are imbalanced. The retrained model is therefore the preferred model for this analysis because it identifies considerably more positive cases and provides a substantially better balance between precision and recall. Nevertheless, the model should be viewed as a decision-support or screening tool rather than a definitive predictor of commercial bestseller status. Future improvements could include optimising the classification threshold, testing different class weights, adding more informative predictors and, most importantly, using an actual sales-based bestseller target if commercial bestseller prediction is required.

Model	Accuracy	Precision	Recall	F1
Original	0.8041082164328658	0.6153846153846154	0.09852216748768473	0.16985138004246284
Retrained	0.753006012024048	0.42245989304812837	0.583743842364532	0.49017580144777667

